

PHY102 Electricity and Magnetism Jan-April 2026: Assignment 1

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To be discussed in the Tutorial on Jan 16

1. Study King's experiment to show the equality of proton charge and electron charge in magnitude. Write 250 words on this experiment in your tutorial notebook.

Note: The Physical Review Letters paper

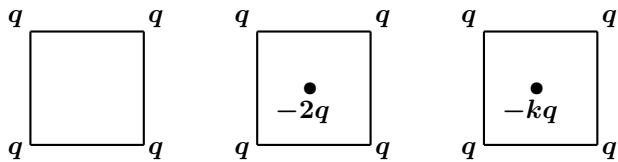
<https://doi.org/10.1103/PhysRevLett.5.562> that describes the experiments has been uploaded to the Moodle page.

2. Study the Cavendish experiment to prove the $1/r^2$ nature of electrostatic forces. Compare this to a more modern version of the Cavendish experiment which limits the variation of the electric force from $1/r^2$ to a very small value. Write 250 words on this experiment in your tutorial notebook.

Note: The Physical Review Letters paper

<https://doi.org/10.1103/PhysRevLett.26.721> that describes the experiments has been uploaded to the Moodle page

3. Calculate the electric energy of the charge configurations in the diagrams (a) (b) (c) below. Take the square side length to be a .



In (c) can there be a value of k for which the energy

is zero? Find that value of k .

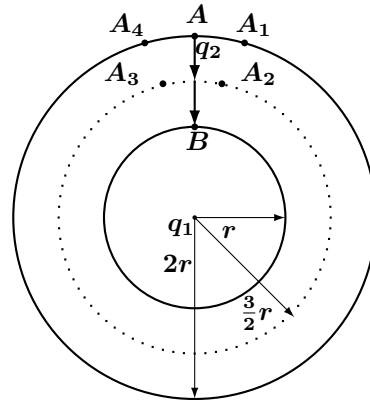
4. Consider bringing a charge q_2 from A to B via three different paths.

a) Path 1: $A \rightarrow B$ (direct radial).

b) Path 2: $A \rightarrow A_1 \rightarrow A_2 \rightarrow B$.

c) Path 3: $A \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow A_4 \rightarrow B$.

Calculate explicitly the work done along each of these paths.



5. Consider two point charges $+q$ at a distance d apart. Calculate the electric field along a line perpendicular to d and drawn at its mid point. As you move along the line does the electric field change direction? Plot the electric field as function of distance along this line for $s = 0$ to $s \rightarrow \infty$.